



Exp-2

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Aim :

To study and understand different types of network cable, To make your own ethernet cross over cable / Straight cable.

Different Types of Network Cable:

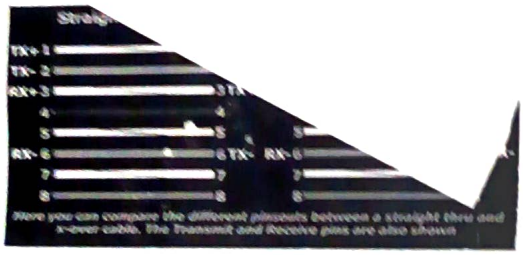
1) Unshielded Twisted Pair Cable
2) Shielded Twisted pair Cable
3) Coaxial Cable
4) Fibre optic Cable.

UTP :

- In Unshield Twisted pair Cable has three categories
- Category 3 has 10bps data Transmission Capacity
- Category 5 has upto 100 mbps data Transmission
- Category 5e has 1Gbps has Capacity.







Here you can compare the different pinouts between a straight thru and x-over cable. The Transmit and Receive pins are also shown

Cable type	Category	Maximum length (Ethernet)	Advantages / Disadvantages	Application
UTP	Category 3	10 Mbps	Advantages • Cheaper in cost • Easy to install	10 Base - T ethernet
	Category 5	Up to 100 Mbps	- they have small overall diameter	Fast ethernet, Gigabit
	Category 5e	1 Gbps	More prone to EMI	
STP	Category 6, 6a	10 Gbps	Advantages • Shielded • Faster • Less prone to noise	• Gigabit ethernet • Widely used in datacenter
Coaxial Cable	RG-6 RG-59 RG-11	10 - 100 Mbps	Advantages • High bandwidth • Immune to interface	• Speed of signal is 500 m • High Speed
Fibre optics Cable	Single mode Multi mode	100 Gbps	Advantages • High Speed • High bandwidth • High Security Disadvantage • Expensive	• Maximum distance of fibre optics Cable is around 100 m

Make your own Ethernet Cross-Over

Tools and parts needed:

- Either Cabling Cat 5e is certified for gigabit, support but Cat 5e Cabling work as well
- A Crimping tool
- Two RJ45 plugs
- Optional two plug shield

Step 1:

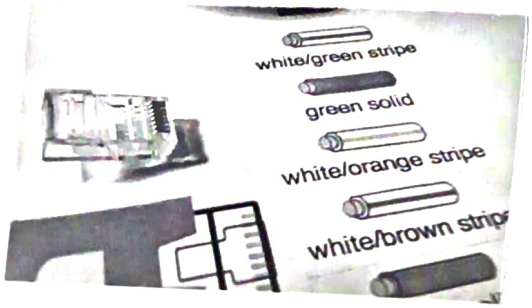
To start construction of the device begin by threading

Step 2:

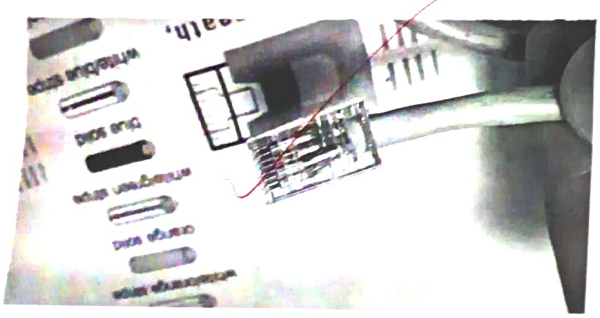
Next strip approximately 1.5 cm of cable shielding from both ends.

Step 3:

After you will need to untangle the wires, there should be four twisted pairs.







Step 4:

Once the order is correct. Bunch them to gather in a line and if there are any that stick out farther than other strip them back to create an even level.

Step 5:

Next push the cable right in. The not chatthendo to be just over the cable shielding and if it isn't that means that you stripped off too much shielding.

Step 6:

After wires are securely sitting inside the plug insert it into crimp - ingo to go and push

Step 7:

Lastly repeat for theother endusing diagram to test it plug it in attempt to connect two devices directly.

Result :

Thus, different types of network cables were studied successfully and straigh through and cross over ethernet cable were tested successfully.